Team Description Paper (TDP)

Team name: RoboKit

Affiliation: Technopark of the P. L. Kapitsa Phystech Lyceum

Country: Russia

City: Dolgoprudny

Team leader: Dmitry Dmitrievich Ustinov

Robot platform: Roki-2 (humanoid robot with 23 servomotors)

Website: https://starkit.su/

Abstract

MechTech RoboKit is a school robotics team based at the humanoid robotics laboratory of the Phystech Lyceum Technopark. Since 2019, we have been assembling, programming and controlling robots using computer vision and modern artificial intelligence tools.

We use Roki-2, a Russian humanoid platform developed at the Moscow Institute of Physics and Technology (MIPT). Students tune motions, develop behaviour algorithms and prepare robots for autonomous soccer and all-round events. Azer Babaev and MIPT's StarKit team support our work.

Competition Participation

RoboCup Asia-Pacific 2019 — 1st and 3rd places; Virtual RoboCup Asia-Pacific 2020 — 1st place. Robofest 2020 — 1st and 2nd in Junior Humanoid Soccer. RoboCup Russia 2021 — victory for Robokit A; FIRA 2021 — 2nd in the overall all-round standings.

RoboScience Hackathon 2023 — 12 podium finishes, including a clean sweep of the triple jump podium.

FIRA RoboWorld Cup and Summit 2024, Brazil — 1st and 2nd in Humanoid Junior. FIRA Korea 2024 — wins in HuroCup All Round, Kids Weightlifting and Marathon. FIRA Eurasia 2024, Türkiye — 1st places in the junior and adult categories.

RoboCup Russia 2024 — wins in autonomous soccer and FIRA overall standings. RoboScience Hackathon 2024 — 21 podium results: nine first, seven second and five third places. Wins: Junior — all-round, basketball, triple jump, marathon and penalty kicks; Pro — all-round, sprint, weightlifting and marathon.

Competition Participation (continued)

RoboCup Russia 2025 — nine podium finishes: three 1st, three 2nd and three 3rd places. The team swept the podium in autonomous soccer and triple jump, also taking 1st and 3rd in Entry and 2nd in sprint.

FIRA 2025, Taiwan — a clean sweep of the school league podium. FIRA RoboWorld Cup 2025, South Korea — 1st in penalty kicks and three 3rd places: triple jump, sprint and penalty kicks.

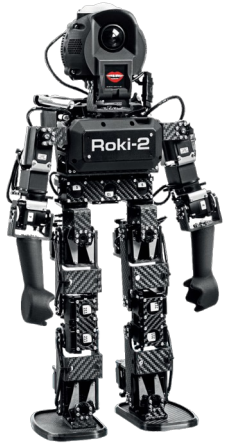
RoboScience Hackathon 2025 — a clean sweep in humanoid soccer, plus 1st and 2nd places in junior all-round.

RoboCup Russia 2026 — six podium results (one first, two second and three third), including the Junior Humanoid Soccer title. FIRA RoboSports Cup Asia, Taiwan — victory in individual HuroCup. Robofinist, 4 October — 1st and 2nd in both autonomous and controlled soccer; Robokit B and Robokit A won respectively.

Related wins: in 2025, Evgeny Kravets was the overall ninth-grade Robofest Olympiad winner; Alexey Yagubov won CoFeRoS Soccer Junior at RoboScience Hackathon. In 2026, Dmitry Ustinov with his team and Ilya Borisov reported ROS Meetup hackathon wins; placings are unspecified in the register.

Hardware Overview

Our main platform is Roki-2, the “Robot Athlete” kit developed at MIPT. In 2025, we also competed with Roki-4 and the Khodok (“Walker”) platform.



The Roki-2 kit includes:

23 servomotors and a preassembled head module

Aluminium and carbon-fibre body parts, together with 3D-printed plastic components

Electronic boards, connecting cables and a battery-powered supply system

Charging equipment, a stand and a transport case

The robot walks, jumps, bends and gets up after falling; it finds coloured objects, follows lines and plays soccer. The development environment runs on an external Windows PC.

Roki-2 platform. Photo: STARKIT, sk24.ru

Software and AI

We extend Roki-2 predefined motions with Python behaviour trees, OpenCV computer vision and controllers for soccer and all-round tasks. Classes include training object-recognition models such as YOLOv8.

Autonomous soccer combines ball search, approach and kicking; marathon tasks use line following. We use Elsiros/Webots for preparation. Neural locomotion control and reinforcement learning remain research directions.

Education and Development

Free weekly classes combine robot assembly, Python, computer vision and teamwork. Students validate their solutions in competitions. Our next priorities are robust autonomous behaviour, recovery after falls and testing in simulation and on real robots.

Results as of 14 October 2026

RoboKit humanoid leagues: 83 podium results — 33 first, 26 second and 24 third places. The wider register, including related achievements, contains 92 records: 35 first, 28 second and 24 third places, 3 third-degree diplomas and 2 wins without a specified placing.

Year	1st	2nd	3rd	Other*	Total
2026	4	4	3	2	13
2025	9	7	8	3	27
2024	17	11	5	0	33
2023	1	4	7	0	12
2021	1	1	0	0	2
2020	2	1	0	0	3
2019	1	0	1	0	2
Total	35	28	24	5	92

* 2025: three third-degree diplomas; 2026: two wins without specified placings. Nine related records are separate from humanoid leagues; Robokit Z’s (2024) affiliation with the laboratory is unconfirmed. Results are not individual medal counts. Unverified 2022 results are excluded.
Full register and sources: dimaystinov.github.io/#results